Doob's Proof of MLE Strong Consistency: A Surprisal and Information Perspective

Omar Hijab
Department of Mathematics, Temple University

2026

Abstract

We give a concise exposition of Doob's proof of strong maximum likelihood consistency, formulated in terms of surprisal and relative information. This formulation clarifies the roles of identifiability, properness, convergence, and continuity, and highlights that consistency holds without uniform convergence of the log-likelihood. We then apply the result to two special cases, i.i.d. sequences and stationary ergodic Markov processes, illustrated by an i.i.d. Gaussian sequence and an autoregressive Gaussian process.

Keywords: maximum likelihood estimator, true parameter, relative information, autoregressive, markov process, law of large numbers

1 Introduction

The consistency of the Fisher (1922) maximum likelihood estimator (MLE) is a classical result discovered more than a century ago. This method is typically treated as a special case of M-estimators, and derived via techniques dating back to Cramér (1946) and Wald (1949). A modern treatment is van der Vaart (1998). Because of its intuitive appeal and broad applicability, the method is a dominant tool of statistical inference.

In teaching this material, I found the Doob (1934) strong consistency proof of MLE convergence more motivated and accessible to students, particularly for students already familiar with information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter spaces and general state spaces with essentially no extra effort only adds to its appeal.

The contribution of this note is expository and pedagogical: We give a formulation of Doob's consistency argument in surprisal/information terminology, in sufficient generality to encompass i.i.d. sequences and Markov processes as special cases. This is done without assuming uniform convergence. To highlight the applicability of our assumptions, we also work out two examples, i.i.d. Gaussian sequences and autoregressive Gaussian processes. We conclude with a brief comparison of MLE and stochastic gradient descent.

2 Surprisal and Information

A random variable x is sampled repeatedly, resulting in samples[1] x_1, x_2, ... We assume the density $p(x, \theta^*)$ of x is one of a parametric family $p(x, \theta)$ of *positive* functions, and we wish to recover θ^* from x_1, x_2, ... While $p(x, \theta)$ may be chosen arbitrarily, based on prior knowledge, we have no control over the underlying expectation E_*. Given this, how do we recover θ^* from the process x_1, x_2, ...?

Define the *surprisal* Samson (1951), Tribus (1961)

$$S(x, \theta) = \log(1/p(x, \theta)).$$

This terminology is intuitive: low probability equals high surprisal.

Assume $S(x, \theta)$ is integrable for all θ. Then it is natural to call a minimizer θ^* of the *expected surprisal* $E_*(S(x, \theta))$ a *true parameter*, and the corresponding density $p(x, \theta^*)$ a *true density*. Equivalently, given θ^* and θ, the *relative information* is

$$I(\theta^*, \theta) = E_*(S(x, \theta)) - E_*(S(x, \theta^*)). \tag{1}$$

The relative information is the excess expected surprisal of θ over θ^*. Then θ^* is a true parameter iff $I(\theta^*, \theta) \geq 0$ for all θ.

To connect this formulation to the standard treatment of MLE consistency, we show that under an additional natural assumption, the true parameter and relative information reduce to their familiar forms.

We say θ^* is a *reference parameter* if

$$E_*\left(\frac{p(x, \theta)}{p(x, \theta^*)}\right) = 1 \qquad \text{for all } \theta.$$

Then the corresponding density $p(x, \theta^*)$ is a *reference density*.

Let θ^* be a reference parameter, and define E and E_θ by

$$E_*(f(x)) = E(p(x, \theta^*) f(x)) \qquad \text{and} \qquad E_\theta(f(x)) = E(p(x, \theta) f(x)).$$

Then $E_\theta(1) = 1$ for all θ and the relative information takes the familiar form Kullback & Leibler (1951)

$$I(\theta^*, \theta) = E\left(p(x, \theta^*) \log\left(\frac{p(x, \theta^*)}{p(x, \theta)}\right)\right). \tag{2}$$

Since $j(p) = \log(1/p)$ is convex, by Jensen's inequality,

$$I(\theta^*, \theta) = E_*\left(j\left(\frac{p(x, \theta)}{p(x, \theta^*)}\right)\right) \geq j\left(E_*\left(\frac{p(x, \theta)}{p(x, \theta^*)}\right)\right) = j(E_\theta(1)) = j(1) = 0, \qquad \text{for all } \theta.$$

[1] We use the same letter (e.g. x) to denote both a random variable and a sample of it.

It follows that a reference density is a true density, and a reference parameter is a true parameter.

If θ is any other true parameter, then $I(\theta^*, \theta) = 0$, hence, by strict convexity of $j(p)$, there is a scalar t such that $p(x, \theta) = t \cdot p(x, \theta^*)$ with probability one.[2] Since θ^* is a reference parameter, $t = 1$, hence $p(x, \theta) = p(x, \theta^*)$ with probability one: every true density equals the reference density. It follows that there is at most one reference density; when there is a reference density, it is the unique true density.

We say parameters θ, θ' *share a density* if $p(x, \theta) = p(x, \theta')$ with probability one. If parameters do not share densities, there is at most one reference parameter; moreover, when there is a reference parameter, it is the unique true parameter.

In the form (2), the relative information $I(\theta^*, \theta)$ is more commonly known as Kullback-Leibler divergence van der Vaart (1998) or relative entropy Cover & Thomas (2006), Dembo & Zeitouni (1998) or both Bishop (2006), Murphy (2022). It is natural to regard $I(\theta^*, \theta)$ as information, following some authors Kullback (1959), Schervish (1995), since information and entropy are conceptual opposites: the Shannon (1948) entropy is concave, while the relative information is convex.

Returning to the general setting, the *likelihood* and (empirical) *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^{n} p(x_k, \theta), \qquad S_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^{n} S(x_k, \theta).$$

The mean surprisal is also called the negative log-likelihood.

By definition, θ^* minimizes the expected surprisal. The MLE method rests on the convergence of the mean surprisal to the expected surprisal for large samples. While this is not the case in general, this convergence is guaranteed in two classical cases: an i.i.d. sequence Fisher (1922), and a stationary ergodic Markov process Anderson & Goodman (1957), Billingsley (1961). In the former case, the convergence is a consequence of the law of large numbers; in the latter case, of the ergodic theorem.

Because of this, it is natural to estimate θ^* by a minimizer of $S_n(\theta)$ or, equivalently, a maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a parameter $\hat{\theta}_n$, depending on the samples, maximizing $L_n(\theta)$ over all θ. In what follows, only E_*, $p(x, \theta)$, and true parameters enter; E, E_θ, and reference parameters play no role.

3 Consistency

In addition to integrability of $S(x, \theta)$, we make four assumptions.

A (Identifiability)**.** *There is at most one true parameter.*

A follows if the expected surprisal is strictly convex. Also, as we saw above, A follows if there is a reference parameter and parameters do not share densities.

A sequence of parameters *subconverges* to θ if a subsequence converges to θ. If a sequence subconverges to θ, θ is a *limit point of the sequence.* A sequence has exactly one limit point θ iff the sequence converges to θ. A subset K of the parameter space is *compact* if every sequence in K has at least one limit point in K.

A function $F(\theta)$ is *proper* if its sublevel sets $F(\theta) \leq c$ are compact subsets of the parameter space, for every level c. Properness prevents minimizing sequences from escaping to infinity. Together with continuity, this forces the existence of a minimizer.

B (Properness)**.** *There is an $N \geq 1$, depending on the samples, and a proper $F(\theta)$, not depending on the samples, such that, with probability one,*

$$F(\theta) \leq S_n(\theta), \qquad \textit{for all } \theta \textit{ and } n \geq N. \tag{3}$$

When the parameter space is compact, the proof below shows assumption B is superfluous.

C (Convergence)**.** *For every integrable $g(x)$*

$$\frac{1}{n} \sum_{k=1}^{n} g(x_k) \to E_*(g(x)), \qquad \textit{as } n \to \infty, \qquad \textit{with probability one.} \tag{4}$$

[2] *With probability one* is always relative to E_*.

This implies (4) with $g(x) = S(x, \theta)$, for every θ. We will need this convergence simultaneously for all θ, in the sense

$$S_n(\theta) \to E_*(S(x,\theta)), \qquad \text{as } n \to \infty, \qquad \textit{for all } \theta\textit{, with probability one.} \tag{5}$$

We emphasize this is not uniform convergence, since the rate of convergence may vary with parameters θ.

D (Separability and Continuity)**.** *There is a countable subset of parameters θ' of the parameter space, a function $\delta(\theta, \theta')$ satisfying $\inf_{\theta'} \delta(\theta, \theta') = 0$ for all θ, and an integrable nonnegative $g(x)$ satisfying*

$$\left| S(x,\theta) - S(x,\hat\theta) \right| \le \delta(\theta,\hat\theta) g(x), \qquad \textit{for all } x \textit{ and } \theta,\ \hat\theta. \tag{6}$$

Under C, D, a standard separability argument establishes (5). Under D, the expected surprisal $E_*(S(x,\theta))$ is continuous in θ. If moreover B, C hold, then by sending $n \to \infty$ in (3), $F(\theta) \le E_*(S(x,\theta))$ for all θ, hence the expected surprisal is proper. Hence there is a unique true parameter under A, B, C, and D. Similarly, under B, D, the mean surprisal is continuous in θ and proper, hence an MLE $\hat\theta_n$ exists for $n \ge N$, with probability one.

These assumptions are particularly tractable for *exponential families* Cover & Thomas (2006), Murphy (2022), Schervish (1995), van der Vaart (1998), in which $S(x,\theta)$ is essentially a sum of products of the form $f(\theta)g(x)$. In §4, we verify these assumptions for an i.i.d. Gaussian sequence and an autoregressive Gaussian process.

In (5), there is no presumption of uniform convergence. This is a key feature of Doob's proof. The argument instead combines properness, convergence (5), and the continuity estimate in (6) to control limit points of the MLE sequence.

Theorem (Consistency)**.** *Let $x_1,\ x_2,\ \ldots$ be a process, and let $p(x,\theta)$ be a parametric family of positive functions, such that the surprisal is integrable for all θ. Under assumptions A, B, C, D,*

1. *there is a unique true parameter θ^*,*
2. *with probability one, an MLE sequence $\hat\theta_n$ exists for sufficiently large n,*
3. *with probability one, any MLE sequence converges to the true parameter, $\hat\theta_n \to \theta^*$.*

Proof. The first two items were derived above. For the third item, with $g(x)$ as in (6), fix samples x_1, x_2, $\ldots$ for which (3), (4), and (5) hold. We show $\hat\theta_n \to \theta^*$ for these samples. By (3),

$$F(\hat\theta_n) \le S_n(\hat\theta_n) \le S_n(\theta^*), \qquad \text{for } n \ge N.$$

By (5), the right side of this last display is bounded as $n \to \infty$. Since $F(\theta)$ is proper, the sequence $\hat\theta_n$ lies in a compact set in the parameter space, and thus has limit points.

Suppose θ is a limit point of $\hat\theta_n$. Then $\hat\theta_n$ subconverges to θ. By (6),

$$S_n(\theta) - S_n(\theta^*) \le S_n(\theta) - S_n(\hat\theta_n) \le \delta(\theta,\hat\theta_n) \cdot \frac{1}{n}\sum_{k=1}^n g(x_k). \tag{7}$$

Since the left side of (4) is bounded as $n \to \infty$, and $\delta(\theta,\hat\theta_n)$ subconverges to zero, the right side of (7) subconverges to zero. By (1) and (5), the left side of (7) converges to $I(\theta^*,\theta)$, hence $I(\theta^*,\theta) \le 0$. But $I(\theta^*,\theta) \ge 0$ since θ^* is a true parameter, hence $I(\theta^*,\theta) = 0$. Since there is only one true parameter, $\theta = \theta^*$. Thus θ^* is the only limit point, hence $\hat\theta_n \to \theta^*$ as $n \to \infty$. □

4 Examples

The first example is MLE for an i.i.d. Gaussian sequence, and the second example is MLE for an autoregressive Gaussian process. In the first example, the sequence is denoted z_1, z_2, $\ldots$—anticipating its role as the innovations driving the autoregressive process of the second example—reserving x_1, x_2, $\ldots$ for that Markov process. Even though the i.i.d. MLE *result* is completely standard, we derive it in detail, both to illustrate our assumptions in this simpler setting and because the same ϵ-perturbation technique is reused in the more involved Markov example below. Throughout $|\cdot|$ denotes the absolute value of a scalar, the euclidean length of a vector, or the euclidean norm of a matrix, as the need arises.

4.1 I.I.D. Gaussian Sequence

Let μ be in $\mathbf{R}^d$, Q a positive definite $d \times d$ matrix, and let $v \cdot w$ be the dot product in $\mathbf{R}^d$. With z in $\mathbf{R}^d$, the *normal density with parameter* $\theta = (\mu, Q)$ is $p(z, \theta) = e^{-S(z,\theta)}$ where

$$S(z, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2}(z - \mu) \cdot Q^{-1}(z - \mu).$$

This defines the parametric family. A random variable z in $\mathbf{R}^d$ is *normally distributed with parameter* $\theta^* = (\mu^*, Q^*)$ if its moment-generating function is

$$E_* \left(e^{v \cdot z}\right) = \exp\left(v \cdot \mu^* + \frac{1}{2} v \cdot Q^* v\right). \tag{8}$$

Let z_1, z_2, ... be independent and identically distributed copies of z. We show $S(z, \theta) = \log(1/p(z, \theta))$ is integrable and θ^* is the true parameter, and we verify A, B, C, and D. Assumption C is verified, as it is the law of large numbers (LLN).

Let $\theta' = (\mu', Q')$, $g(z) = 1 + |z|^2$ and let $\delta(\theta, \theta')$ equal half of

$$|\log \det Q - \log \det Q'| + \left|Q^{-1}\mu - (Q')^{-1}\mu'\right| + |Q^{-1} - (Q')^{-1}| + \left|\mu \cdot Q^{-1}\mu - \mu' \cdot (Q')^{-1}\mu'\right|.$$

Then $\delta(\theta, \theta')$ is continuous. By standard manipulations (6) follows, establishing D.

Let A and B be symmetric matrices. Throughout we say $A \geq B$ if $v \cdot Av \geq v \cdot Bv$ for all vectors v. Then the trace is monotone in the sense $A \geq B$ and $Q > 0$ imply $\operatorname{trace}(Q^{-1}A) \geq \operatorname{trace}(Q^{-1}B)$.

Proposition. *For $Q^* > 0$ and c scalar, let $\alpha = e^{-2c-1} \det(2\pi e Q^*)$ and*

$$F_*(Q, Q^*) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} \operatorname{trace}(Q^{-1}Q^*), \qquad Q > 0.$$

Then $F_*(Q, Q^*)$ is minimized iff $Q = Q^*$, and $F_*(Q, Q^*) \leq c$ implies $\alpha Q^* \leq Q \leq Q^*/\alpha$.

Proof. Let $h(\lambda) = \log \lambda + 1/\lambda$, $\lambda > 0$. Then $h(\lambda)$ is minimized iff $\lambda = 1$ and $h(\lambda)$ is proper in the sense $h(\lambda) \leq \log(1/\alpha)$ implies $\alpha \leq \lambda \leq 1/\alpha$. Write $Q^* = S^2$ with $S > 0$, and define Z by $Q = SZS$. Suppose $F_*(Q, Q^*) \leq c$ and let λ_1, λ_2, ..., λ_d be the eigenvalues of Z. Then the proposition follows from

$$\log(1/\alpha) - 1 \geq 2F_*(Q, Q^*) - 2F_*(Q^*, Q^*) = \log \det Z + \operatorname{trace}(Z^{-1} - I) = \sum_{i=1}^{d} (h(\lambda_i) - 1). \qquad \square$$

By monotonicity of the trace, $F_*(Q, Q^*)$ is analogously monotone in Q^*. Given vectors $v = (s_1, s_2, \ldots, s_d)$ and $w = (t_1, t_2, \ldots, t_d)$, the *tensor product* $v \otimes w$ is the matrix whose ij-th entry is $s_i t_j$. Then $z \cdot Q^{-1} z = \operatorname{trace}(Q^{-1}(z \otimes z))$. Using the moment generating function, it follows $S(x, \theta)$ is integrable and

$$E_*(S(z, \theta)) = F_*(Q, Q^*) + \frac{1}{2}(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu).$$

for each θ. By the proposition, (μ^*, Q^*) is the unique minimizer of the expected surprisal, establishing A.

Now let $0 < \epsilon < 1/2$ and

$$F_\epsilon(\theta) = F_*(Q, (1 - 2\epsilon)Q^*) + \frac{1}{2}(1 - \epsilon)(\mu^* - \mu) \cdot Q^{-1}(\mu^* - \mu), \qquad \theta = (\mu, Q).$$

By the proposition with $(1 - 2\epsilon)Q^*$ replacing Q^*, if $F_\epsilon(\theta) \leq c$, then Q is confined to a compact subset of positive definite matrices and, subsequently, μ is confined to a compact subset of $\mathbf{R}^d$. Hence $F_\epsilon(\theta)$ is proper.

With

$$Q_n(\mu) = \frac{1}{n} \sum_{k=1}^{n} (z_k - \mu) \otimes (z_k - \mu), \qquad \hat{\mu}_n = \frac{1}{n} \sum_{k=1}^{n} z_k, \qquad \hat{Q}_n = Q_n(\hat{\mu}_n),$$

let $\hat{\theta}_n = (\hat{\mu}_n, \hat{Q}_n)$. Then $\hat{\mu}_n$ and $\hat{Q}_n$ are the usual sample mean and the biased sample covariance.

By the law of large numbers, with probability one, the matrix sequences

$$(\mu^* - \hat{\mu}_n) \otimes (\mu^* - \hat{\mu}_n) \to 0, \qquad Q^* - Q_n(\mu^*) \to 0, \qquad \text{as } n \to \infty.$$

If Q_n is any sequence of symmetric matrices with $Q_n \to 0$, then $|Q_n| \to 0$. Since $Q_n \le |Q_n| I$, given any $Q > 0$, there is $N \ge 1$ such that $Q_n \le Q$ for $n \ge N$. By choosing $Q = \epsilon^2 Q^*$ and $Q = \epsilon Q^*$, it follows there is $N \ge 1$, depending on the samples, such that, with probability one,

$$(\mu^* - \hat{\mu}_n) \otimes (\mu^* - \hat{\mu}_n) \le \epsilon^2 Q^*, \qquad Q_n(\mu^*) \ge (1-\epsilon) Q^*, \qquad \text{for } n \ge N. \tag{9}$$

The cutoff N depends on θ^* but not on θ. Fix samples $x_1, x_2, \dots$ for which (9) holds. We establish (3) for these samples and $F(\theta) = F_\epsilon(\theta)$. Let

$$f_0(\mu) = Q^* + (\mu^* - \mu) \otimes (\mu^* - \mu), \qquad f_\epsilon(\mu) = (1-\epsilon) f_0(\mu) - \epsilon Q^*.$$

Then $F_\epsilon(\theta) = F_*(Q, f_\epsilon(\mu))$.

Insert $L = I$ and $W = w = \hat{\mu}_n - \mu^*$ and $V = v = \mu^* - \mu$ (as single-column matrices) in the matrix ϵ-Young's inequality,[3]

$$\pm (VW^t + WV^t) \le \epsilon V L V^t + \frac{1}{\epsilon} W L^{-1} W^t. \tag{10}$$

This leads to

$$-(v \otimes w + w \otimes v) \le \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w \le \epsilon f_0(\mu), \qquad \text{for all } \mu \text{ and } n \ge N.$$

Writing $z_k - \mu = (z_k - \mu^*) + v$, we have $Q_n(\mu) = Q_n(\mu^*) + (v \otimes w + w \otimes v) + v \otimes v$. Combined with the above inequalities, this leads to

$$Q_n(\mu) \ge f_\epsilon(\mu), \qquad \text{for all } \mu \text{ and } n \ge N.$$

In particular, since $f_\epsilon(\mu) \ge (1-2\epsilon) Q^*$, we have shown $\hat{Q}_n > 0$ for $n \ge N$.

Since $S_n(\theta) = F_*(Q, Q_n(\mu))$ and $Q_n(\mu) \ge \hat{Q}_n$ by least squares, by the proposition and monotonicity of $F_*(Q, Q^*)$ in Q^*,

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, \hat{Q}_n) \ge F_*(\hat{Q}_n, \hat{Q}_n) = S_n(\hat{\theta}_n), \qquad \text{for all } \theta \text{ and } n \ge N.$$

Thus $\hat{\theta}_n$ is the MLE for $n \ge N$, and

$$S_n(\theta) = F_*(Q, Q_n(\mu)) \ge F_*(Q, f_\epsilon(\mu)) = F_\epsilon(\theta), \qquad \text{for all } \theta \text{ and } n \ge N.$$

This establishes (3) and completes the derivation of B.

4.2 Autoregressive Gaussian Process

Let A be a $d \times d$ matrix. With x_0, x_1 in $\mathbf{R}^d$, the *transition density with parameter* $\theta = (\mu, A, Q)$ is $p(x_0, x_1, \theta) = e^{-S(x_0, x_1, \theta)}$ where

$$S(x_0, x_1, \theta) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2} (x_1 - Ax_0 - \mu) \cdot Q^{-1} (x_1 - Ax_0 - \mu).$$

This defines the parametric family.

Let $\theta^* = (\mu^*, A^*, Q^*)$, and assume A^* is a stable matrix. Define

$$c^* = (I - A^*)^{-1} \mu^*, \qquad L^* = \sum_{n=0}^{\infty} (A^*)^n Q^* ((A^*)^t)^n, \tag{11}$$

both well-defined by stability of A^*; as shown below, these are the values of the mean and covariance of x_0 needed to make $x_1, x_2, \dots$ (defined via the recurrence (12) below) stationary.

[3]proof: with $L = S^2$, $S > 0$, and $A = \sqrt{\epsilon} VS \pm WS^{-1}/\sqrt{\epsilon}$, write out $AA^t \ge 0$.

Let z be normally distributed with parameter (μ^*, Q^*), and let x_0 be normally distributed with parameter (c^*, L^*). The *innovations process* $z_1, z_2, \ldots$ consists of i.i.d. copies of z, assumed independent of x_0. The *autoregressive Gaussian process (of order one), with parameter* $\theta^* = (\mu^*, A^*, Q^*)$, is the process $x_1, x_2, \ldots$ generated by the recurrence

$$x_n = A^* x_{n-1} + z_n, \qquad n \geq 1. \tag{12}$$

Then z_n is independent of $x_1, \ldots, x_{n-1}$, and $x_1, x_2, \ldots$ is a Markov Gaussian process.

Let $(x_0, x_1), (x_1, x_2), \ldots,$ be the *pair process*. We show $S(x_0, x_1, \theta)$ is integrable and θ^* is the true parameter, and we verify A, B, C, and D for the pair process, assuming stability of A^*.

For the pair process, the likelihood and mean surprisal are

$$L_n(\theta) = \prod_{k=1}^{n} p(x_{k-1}, x_k, \theta), \qquad S_n(\theta) = \frac{1}{n} \log(1/L_n(\theta)) = \frac{1}{n} \sum_{k=1}^{n} S(x_{k-1}, x_k, \theta).$$

We verify C first. Because the series in (11) converges, we can go backward in time and present x_0 more explicitly. Let $z_0, z_{-1}, z_{-2}, \ldots$ be an i.i.d. Gaussian sequence, with parameter (μ^*, Q^*), independent of $z_1, z_2, \ldots$. Then $\ldots, z_{-2}, z_{-1}, z_0, z_1, z_2, \ldots$ is the *extended* innovations process. If we set

$$x_0 \equiv \sum_{k=0}^{\infty} (A^*)^k z_{-k},$$

then by (11) x_0 is normally distributed with parameter (c^*, L^*) and independent of $z_1, z_2, \ldots$. Moreover iterating (12) and combining,

$$x_n = (A^*)^n x_0 + \sum_{k=1}^{n} (A^*)^{n-k} z_k = \sum_{k=-\infty}^{n} (A^*)^{n-k} z_k, \qquad n \geq 1. \tag{13}$$

This exhibits $x_1, x_2, \ldots$ as a shift-invariant function of the extended innovations process. Since the extended innovations process is i.i.d., it is stationary and ergodic, hence $x_1, x_2, \ldots$ is stationary and ergodic. By the ergodic theorem Birkhoff (1931), Billingsley (1961), for every integrable function $g(x_0, x_1)$,

$$\frac{1}{n} \sum_{k=1}^{n} g(x_{k-1}, x_k) \to E_*(g(x_0, x_1)), \qquad \text{as } n \to \infty, \qquad \textit{with probability one.} \tag{14}$$

This establishes C for the pair process.

For D, with $\theta' = (\mu', A', Q')$, replace the previous $g(z)$ by $g(x_0, x_1) = 1 + |x_0|^2 + |x_1|^2$, and augment the previous $\delta(\theta, \theta')$ by three additional terms, equal to half of

$$|Q^{-1}A - (Q')^{-1}A'| + |A^t Q^{-1}\mu - (A')^t (Q')^{-1}\mu'| + |A^t Q^{-1} A - (A')^t (Q')^{-1}(A')|.$$

This establishes D.

Let $D = A^* - A$ be the difference and let $v = (\mu^* - \mu) + Dc^*$. The mean and variance of

$$x_k - Ax_{k-1} - \mu = z_k + Dx_{k-1} - \mu = (z_k - \mu^*) + D(x_{k-1} - c^*) + v \tag{15}$$

are v and $Q^* + DL^*D^t$. Here we used $c^* - A^*c^* = \mu^*$ (11). If

$$f_0(\mu, A) = Q^* + DL^*D^t + v \otimes v,$$

we conclude

$$E_*(S(x_0, x_1, \theta)) = F_*(Q, f_0(\mu, A)) \equiv F_0(\mu, A, Q)$$

is the sum of three terms. Since the first term is $F_*(Q, Q^*)$, $Q = Q^*$ is its unique minimizer. Since $L^* > 0$, $A = A^*$ is the unique minimizer of the second term, hence $(\mu, A) = (\mu^*, A^*)$ is the unique minimizer of the sum of the second and third terms. This establishes A. Moreover, since the second and third terms are nonnegative, inspecting $F_0(\mu, A, Q) \leq c$ implies Q, then A, then μ are confined to compact subsets of their domains, hence $F_0(\mu, A, Q)$ is proper.

Our last item is establishing B. With the previously defined $\hat{\mu}_n$, here we also need

$$\hat{c}_n = \frac{1}{n}\sum_{k=1}^n x_{k-1}, \qquad \hat{L}_n = \frac{1}{n}\sum_{k=1}^n (x_{k-1} - c^*) \otimes (x_{k-1} - c^*), \qquad \hat{C}_n = \frac{1}{n}\sum_{k=1}^n (z_k - \mu^*) \otimes (x_{k-1} - c^*).$$

Let

$$Q_n(\mu, A) = \frac{1}{n}\sum_{k=1}^n (x_k - Ax_{k-1} - \mu) \otimes (x_k - Ax_{k-1} - \mu).$$

Then the previously defined $Q_n(\mu)$ equals $Q_n(\mu, A^*)$ and $S_n(\theta) = F_*(Q, Q_n(\mu, A))$.

By (15), with $w = \hat{\mu}_n - \mu^*$ and $u = D(\hat{c}_n - c^*)$, the trinomial expansion of $Q_n(\mu, A)$ is the sum of three square terms and three cross terms,

$$Q_n(\mu, A) = Q_n(\mu^*, A^*) + D\hat{L}_n D^t + v \otimes v + (u \otimes v + v \otimes u) + (w \otimes v + v \otimes w) + (D\hat{C}_n^t + \hat{C}_n D^t).$$

For $0 < \epsilon < 1/3$, set

$$f_\epsilon(\mu, A) = (1 - 3\epsilon)Q^* + (1 - 3\epsilon)DL^*D^t + (1 - 2\epsilon)v \otimes v,$$

and $F_\epsilon(\theta) = F_*(Q, f_\epsilon(\mu, A))$. Then, as before, $F_\epsilon(\theta)$ is proper.

By C, with probability one, $Q_n(\mu^*, A^*)$, $\hat{L}_n$, $\hat{c}_n$, $\hat{\mu}_n$, and $\hat{C}_n$ converge to Q^*, L^*, c^*, μ^*, and $E_*((z_1 - \mu^*) \otimes (x_0 - c^*))$ respectively. By independence of z_1 and x_0, the last limit vanishes, thus $\hat{C}_n \to 0$ with probability one. Thus there is $N \geq 1$, depending on the samples, such that, with probability one,

$$\begin{matrix} Q_n(\mu^*, A^*) \geq (1-\epsilon)Q^*, \qquad \hat{L}_n \geq (1-\epsilon)L^*, \qquad (\hat{c}_n - c^*) \otimes (\hat{c}_n - c^*) \leq \epsilon^2 L^*, \\ (\hat{\mu}_n - \mu^*) \otimes (\hat{\mu}_n - \mu^*) \leq \epsilon^2 Q^*, \qquad \hat{C}_n (L^*)^{-1} \hat{C}_n^t \leq \epsilon^2 Q^*, \end{matrix} \qquad \text{for } n \geq N.$$

The cutoff N depends on θ^* but not on θ. Fix samples $x_1, x_2, \dots$ for which these hold. We establish (3) for these samples and $F(\theta) = F_\epsilon(\theta)$.

Using (10), the negative of the sum of the cross terms is bounded above by $2\epsilon f_0(\mu, A)$ for $n \geq N$, hence

$$Q_n(\mu, A) \geq f_\epsilon(\mu, A), \qquad \text{for all } \mu, A \text{ and } n \geq N.$$

Since $S_n(\theta) = F_*(Q, Q_n(\mu, A))$, (3) follows. This establishes B.

The MLE $\hat{\theta}_n$ is obtained by doing least squares of $Q_n(\mu, A)$ over (μ, A). This returns the (multivariate) Yule–Walker estimator Yule (1927), Walker (1931).

5 Concluding Remarks

In machine learning, neural networks are trained to have input-output behavior close to sampled reference behavior $x_1, x_2, \dots$. The goal is to recover the weights w^* minimizing the expected error $J(w) = E_*(S(x, w))$, where the expectation is over the entire dataset. Traditionally, such a minimization is carried out by following the gradient of the expected error. Because the dataset is large, this is not feasible and the gradient of the sampled error $S(x_k, w)$ is followed instead. Thus $\nabla S(x_k, w)$ is computed instead of $\nabla J(w)$. Under appropriate tuning, this minimization procedure converges to the minimum of the expected error. This is stochastic gradient descent (SGD) Robbins & Monro (1951), Hijab (2026). This suggests that MLE and SGD are two complementary strategies for similar minimization problems, allowing the techniques of each method to inform the other.

6 Disclosures

The author reports no funding was obtained for this work and there are no competing interests to declare. The author used a generative AI tool (Anthropic's *Claude Sonnet*) to assist with language editing and bibliographic searches. All AI-assisted content was reviewed, verified, and edited by the author, who takes full responsibility for the content, accuracy, and originality of this work.

References

Anderson, T. W. & Goodman, L. A. (1957), 'Statistical inference about markov chains', *Annals of Mathematical Statistics* **28**(1), 89–110.

Billingsley, P. (1961), *Statistical Inference for Markov Processes*, University of Chicago Press, Chicago.

Birkhoff, G. D. (1931), 'Proof of the ergodic theorem', *Proceedings of the National Academy of Sciences* **17**(12), 656–660.

Bishop, C. M. (2006), *Pattern Recognition and Machine Learning*, Information Science and Statistics, Springer, New York.

Cover, T. M. & Thomas, J. A. (2006), *Elements of Information Theory*, 2nd edn, Wiley-Interscience.

Cramér, H. (1946), *Mathematical Methods of Statistics*, Princeton University Press.

Dembo, A. & Zeitouni, O. (1998), *Large Deviations Techniques and Applications*, Vol. 38 of *Stochastic Modelling and Applied Probability*, 2nd edn, Springer, New York.

Doob, J. L. (1934), 'Probability and statistics', *Transactions of the American Mathematical Society* **36**, 759–775.

Fisher, R. A. (1922), 'On the mathematical foundations of theoretical statistics', *Philosophical Transactions of the Royal Society of London, Series A* **222**, 309–368.

Hijab, O. (2026), *Math for Data Science*, Springer, Cham. Corrected printing.

Kullback, S. (1959), *Information Theory and Statistics*, Wiley, New York.

Kullback, S. & Leibler, R. A. (1951), 'On information and sufficiency', *Annals of Mathematical Statistics* **22**(1), 79–86.

Murphy, K. P. (2022), *Probabilistic Machine Learning: An Introduction*, MIT Press.

Robbins, H. & Monro, S. (1951), 'A stochastic approximation method', *The Annals of Mathematical Statistics* **22**(3), 400–407.

Samson, E. W. (1951), Fundamental natural concepts of information theory, Report E5079, Communications Laboratory, Electronics Research Division, Air Force Cambridge Research Center, Cambridge, MA.

Schervish, M. J. (1995), *Theory of Statistics*, Springer, New York.

Shannon, C. E. (1948), 'A mathematical theory of communication', *Bell System Technical Journal* **27**(3), 379–423.

Tribus, M. (1961), *Thermostatics and Thermodynamics: An Introduction to Energy, Information and States of Matter, With Engineering Applications*, D. Van Nostrand Company, Princeton, NJ.

van der Vaart, A. W. (1998), *Asymptotic Statistics*, Vol. 3 of *Cambridge Series in Statistical and Probabilistic Mathematics*, Cambridge University Press, Cambridge.

Wald, A. (1949), 'Note on the consistency of the maximum likelihood estimate', *Annals of Mathematical Statistics* **20**(4), 595–601.

Walker, G. (1931), 'On periodicity in series of related terms', *Proceedings of the Royal Society of London, Series A* **131**(818), 518–532.

Yule, G. U. (1927), 'On a method of investigating periodicities in disturbed series, with special reference to wolfer's sunspot numbers', *Philosophical Transactions of the Royal Society of London, Series A* **226**, 267–298.